Kathmandu University

Department
of
Electrical and Electronics Engineering

Design of a LEO CubeSat Constellation For Remote Monitoring

Final Year Project Progress Presentation

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Outline of the Presentation

- Background and Introduction
- Problem Definition
- Objective of the Project
- Project Methodology
- Simulation and Results
- Work Accomplishment
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Background and Introduction

- Nepal ranks 4th globally in climate vulnerability, with glacial lakes, landslides, and floods posing constant threats and limited monitoring.
- A low-cost LEO CubeSat constellation has potential to deliver environmental surveillance across all of Nepal's terrain.
- Moving from ground-based or sparse satellite data to a dedicated space-based monitoring network to improve early warning reliability.
- A complete constellation plan with satellite count, orbit configuration and its evaluation and data coverage maps for Nepal and its interconnection with ground stations.

Problem Definition

- LEO CubeSats give short orbital periods, so frequent passes and fast revisit.
- Nepal's terrain hosts glacial lakes prone to GLOFs and other after effects of the glacier lake outburst.
- Ground sensors are costly and often absent in the most endangered regions.
- A satellite constellation can have a good average revisit and still leave hours of the day uncovered because passes cluster.
- For the project, our approach is to treat orbit selection as a black-box optimisation over a full-fidelity simulation.

Objectives

The objectives of the project are:

- To formulate LEO orbital parameters as a Bayesian Optimization problem for single-satellite and incrementally grown multi-satellite, multi-plane constellations, maximizing Nepal coverage while minimizing revisit time for six at-risk glacial lakes.
- To develop temporal-complementarity criteria using Wasserstein distance, largest-gap, and gap-uniformity measures to promote narrow and evenly distributed daily revisit intervals.
- To verify coverage and revisit performance through independent geometric checks and per-lake analysis, establishing day-and-night monitoring capability for Nepal's highest-risk glacial lakes.

Project Methodology: Simulation Pipeline

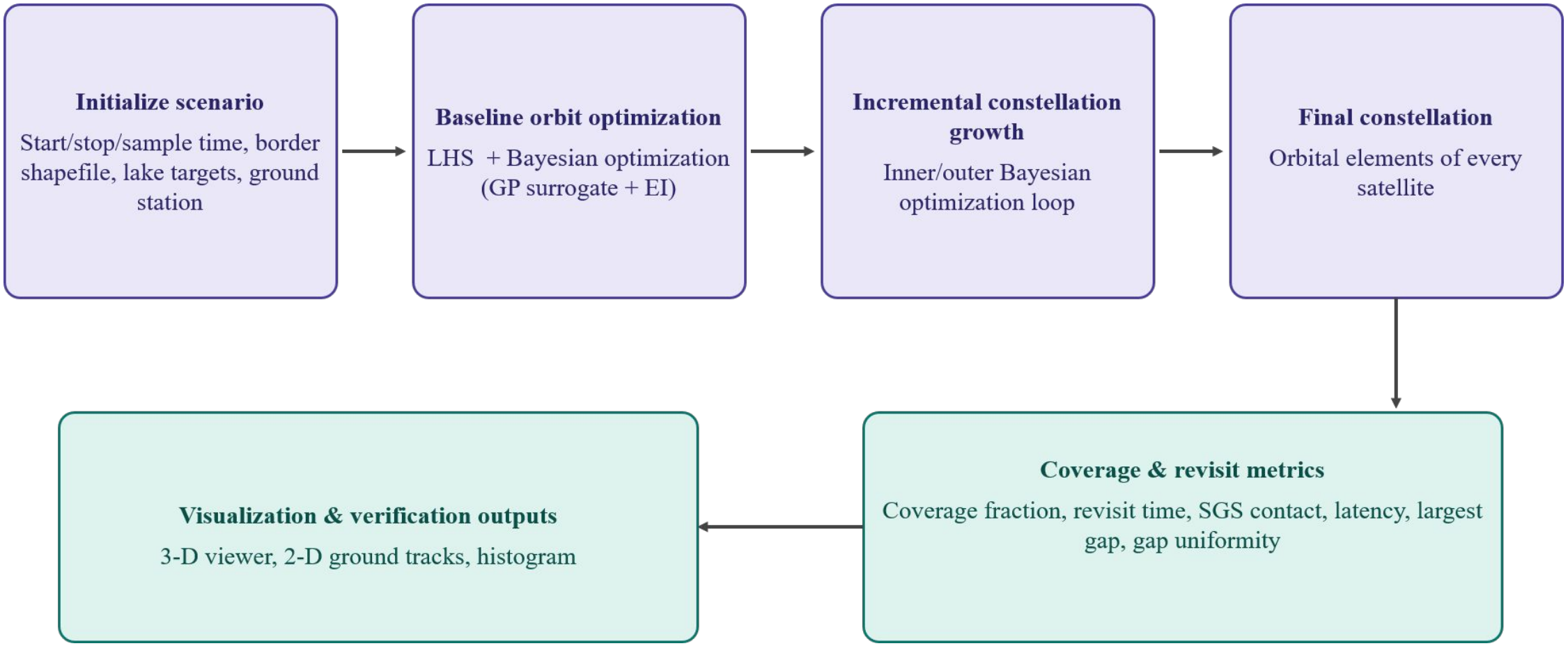


Figure 1: Orbital Design Simulation Architecture

Project Methodology: Single-Satellite Optimisation

- Search space: inclination 0-90°, RAAN 0-360°, true anomaly 0-360° with a height 550 km.
- 100 LHS seed samples spread over the space.
- GP surrogate fitted to all scored candidates.
- EI acquisition proposes the next orbit; 100 further iterations.
- Mission score: coverage, revisit, SGS contact, latency, lake coverage, each normalised and weighted.

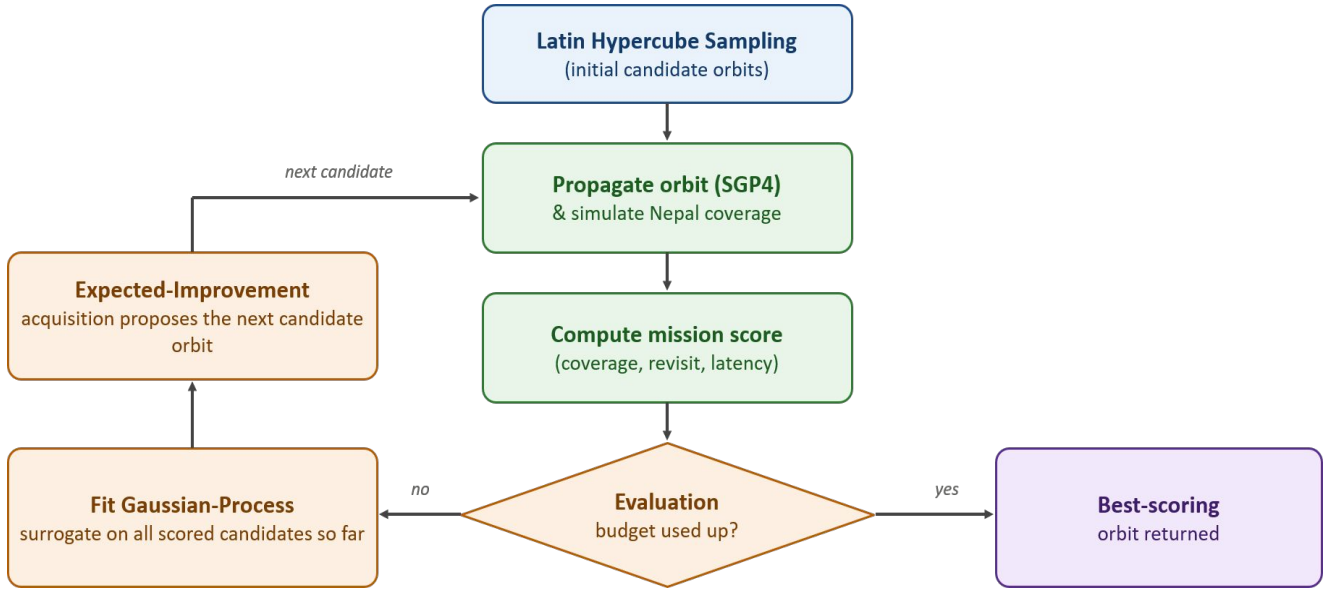


Figure 2: Bayesian Optimization (Latin Hypercube Sampling + Gaussian Process Surrogate + Expected Improvement)

Why BO? Each evaluation is a full propagate-and-simulate run, so a sample-efficient search beats a grid or an analytic approximation.

Project Methodology: Incremental Constellation Growth

- Inner BO: True anomaly and altitude offset (±40 km, ≥ 15 km from co-plane satellites).
- Outer BO: Inclination and RAAN of a new plane, with a nested search for its first satellite.
- Safeguards: Accept on a coverage, revisit, latency or gap gain and hard veto implemented to satisfy condition: worst-lake revisit may not regress by > 15 min.

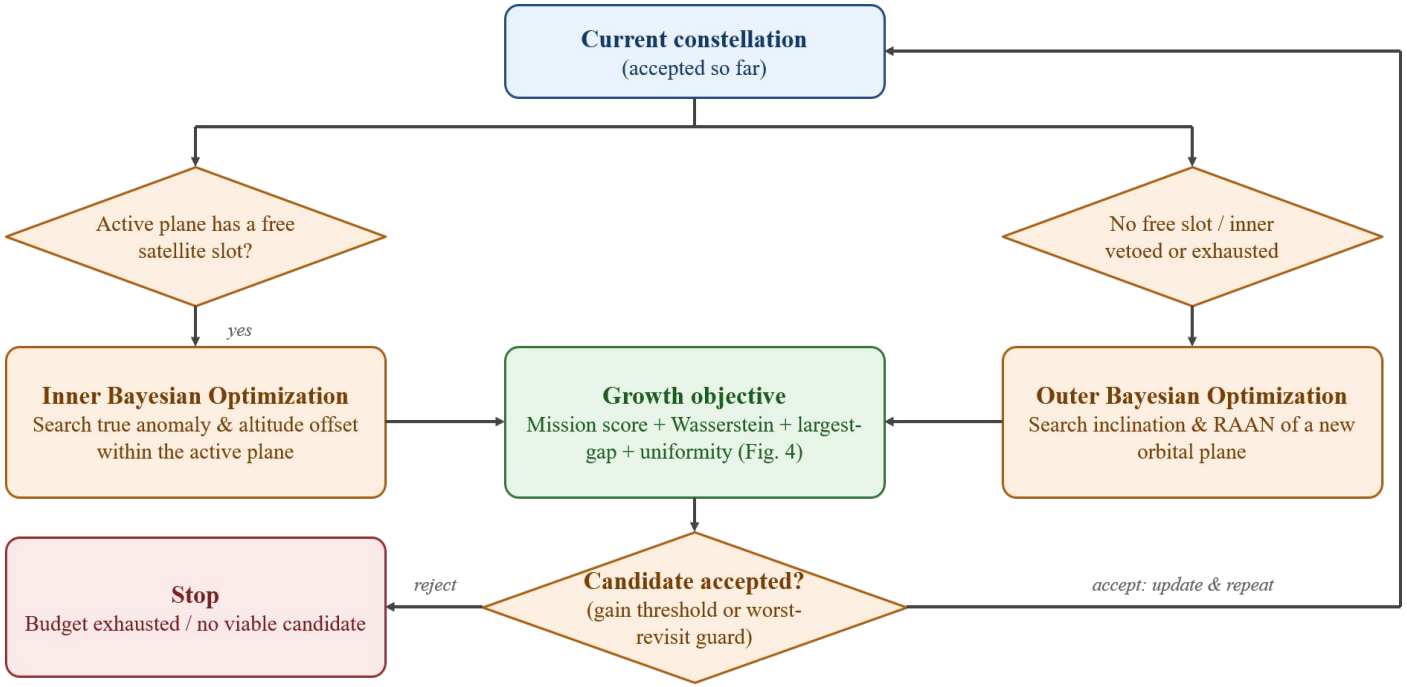


Figure 3: Incremental Constellation Growth Loop

Specified Budget (Set Hypothesis): maximum number of orbital planes are 3 and maximum number of satellite per plane is also 3.

Project Methodology: Growth Objective

- Wasserstein distance: Favors satellites whose passes occur at different times, adding new coverage periods rather than duplicating existing coverage.
- Largest pass-time gap: Favors satellites that reduce the longest gap between consecutive passes, including gaps across midnight.
- Gap uniformity: Favors satellites that spread passes more evenly throughout the day by reducing the overall size of coverage gaps.

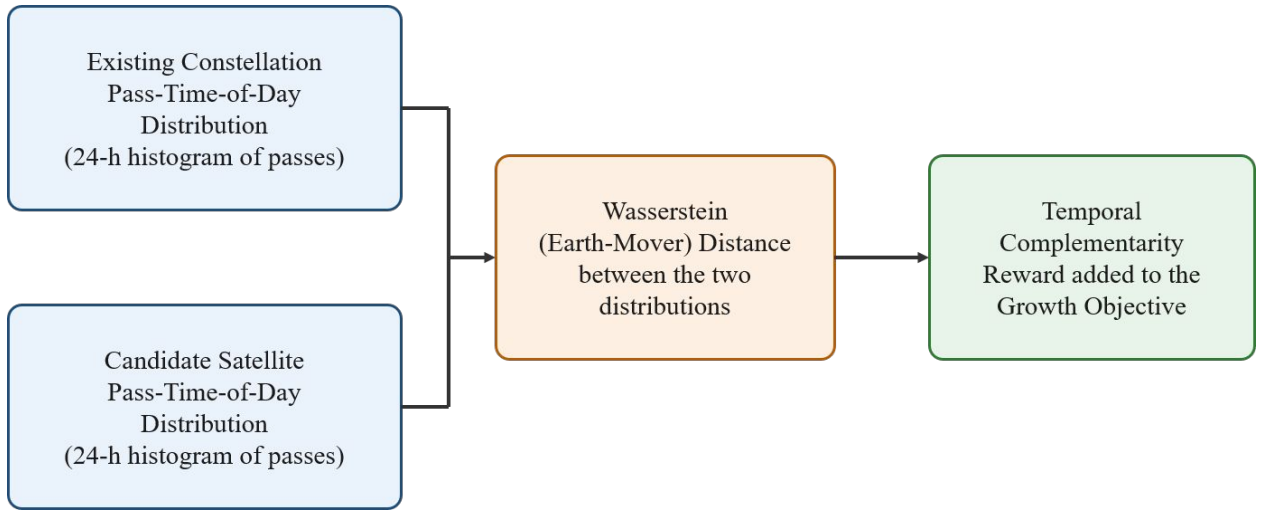


Figure 4: Wasserstein-Distance Temporal-Complementarity Term

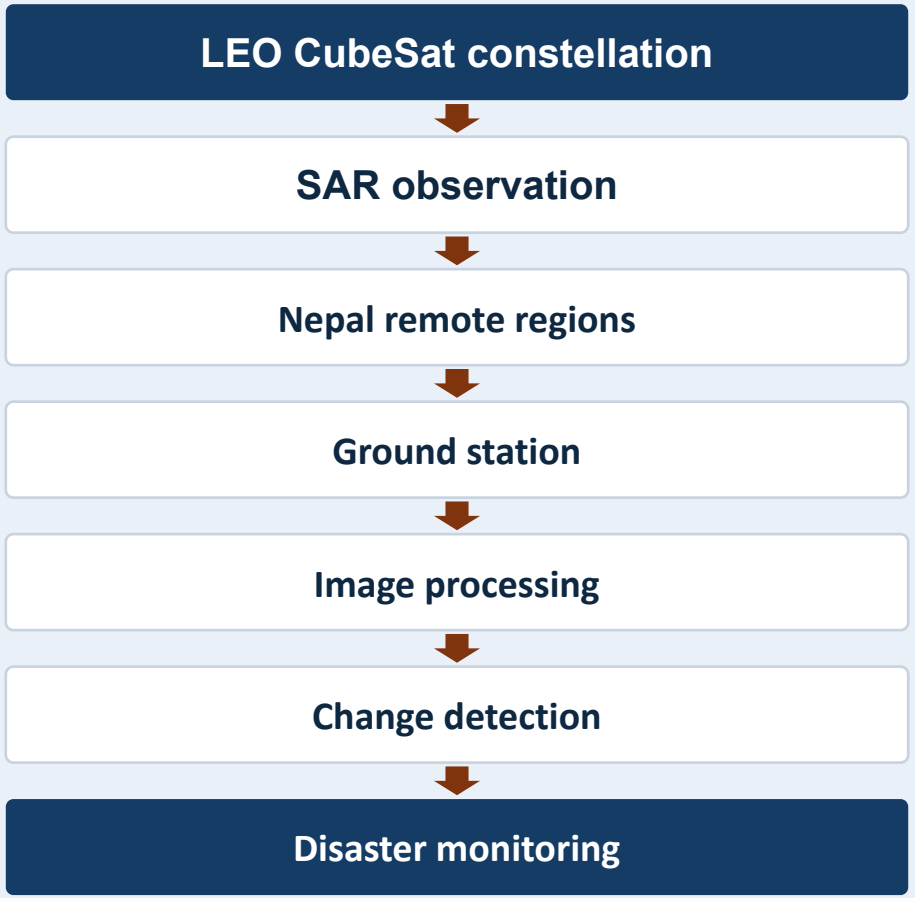
Growth objective: A measure that rewards how much a new satellite improves the existing constellation's overall performance or coverage.

Project Methodology: Remote Monitoring

Why monitor from space?

- Remote mountain regions are hard to monitor from the ground
- Our CubeSat constellation observes them again and again
- A SAR payload sees day and night, even through cloud
- Equips each CubeSat with a fix 15° conical half angle to simulate Earth-observation payloads.

How the system works



Remote Monitoring: Why SAR for Remote Monitoring?

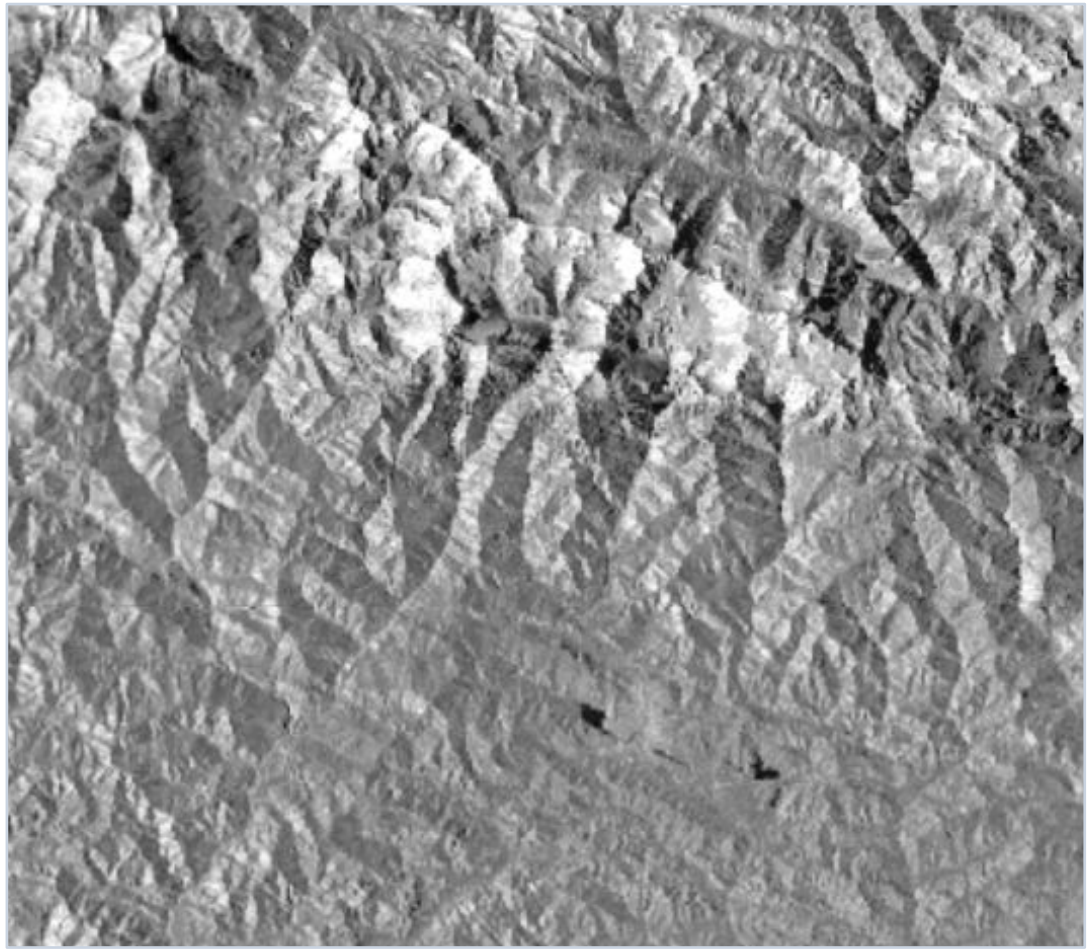


Figure 5: Sentinel-1 SAR image of mountainous terrain, from Google Earth Engine

Optical sensor

- Needs sunlight
- Blocked by clouds

SAR (radar)

- Day and night
- Sees through cloud

Why it suits Nepal?

- Works over mountainous, often cloudy regions
- Detects water, useful for floods and lakes
- Repeat images reveal change over time
- Used for observe change in behavior of glacier lake

Remote Monitoring: Imja Tsho

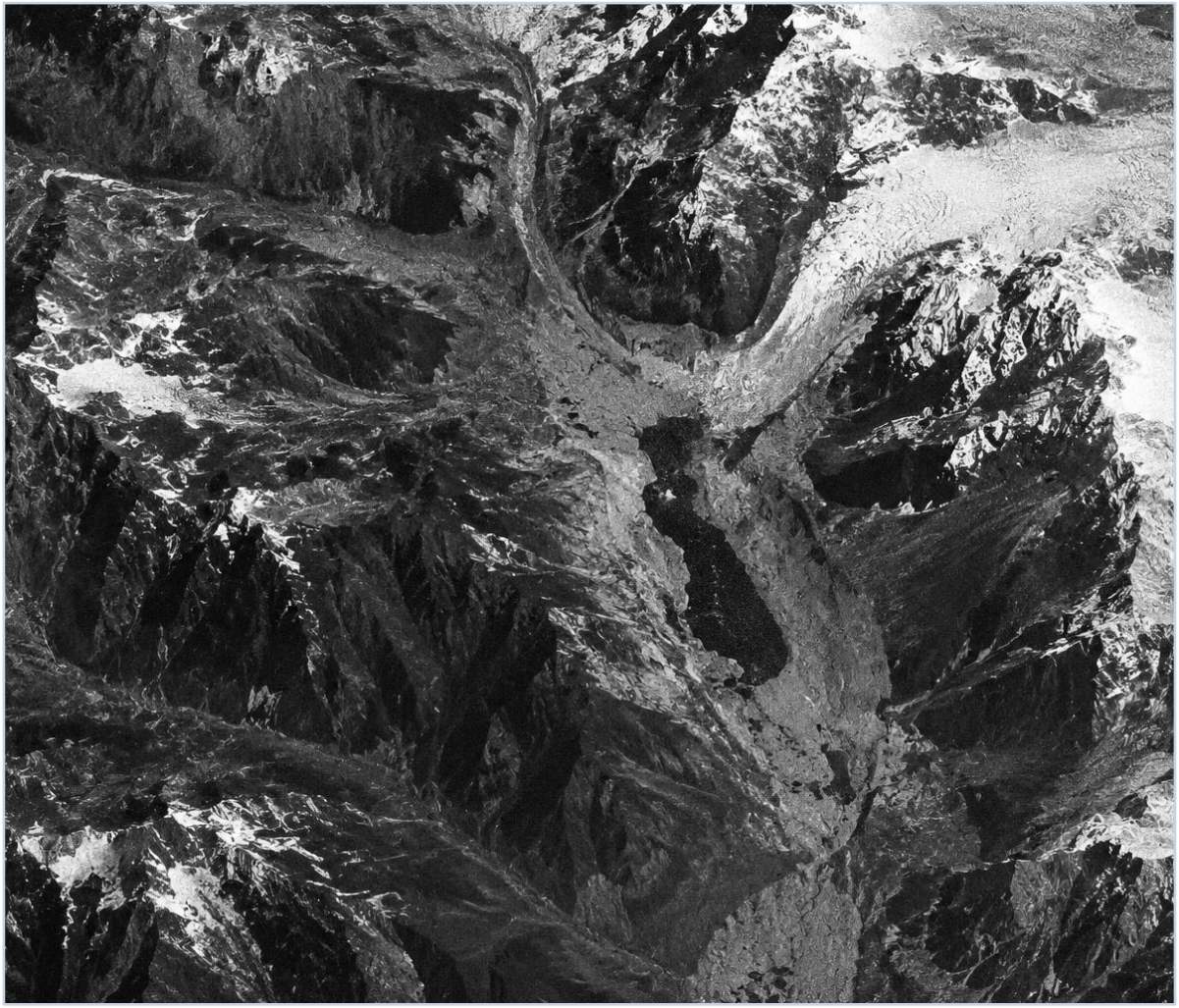


Figure 6: Sentinel-1 SAR image of the Imja Tsho region, from Google Earth Engine

Monitoring approach

1. Observe the lake with SAR
2. Observe again on a later date
3. Compare the two images
4. Detect change in lake or water area

Target: Imja Tsho, Nepal
Data: Sentinel-1 SAR
Platform: Google Earth Engine

Sentinel-1 is not our CubeSat.
Its real SAR data stands in for what our CubeSat payload could observe.

From Constellation to Remote Monitoring

MATLAB: when can we observe?

- Satellite position
- Ground track
- Visibility of the target
- Imaging opportunity
- Observation time

Earth Engine: what changed?

- Sentinel-1 SAR data
- SAR image
- Image processing
- Change detection

GLOF Observation and Monitoring

Conclusion

A CubeSat constellation that repeatedly observes remote regions of Nepal with SAR, using those observations to detect changes related to GLOFs, floods and the environment.

Simulation and Result: Best Single-Satellite Orbit

28.66° inclination	**99.47 min** average revisit
85.11% coverage fraction	**3** passes per day
77.33 min SGS contact (7 contacts)	**94.33 min** downlink latency

RAAN 345.45° · true anomaly 32.74° · mission score 0.0347

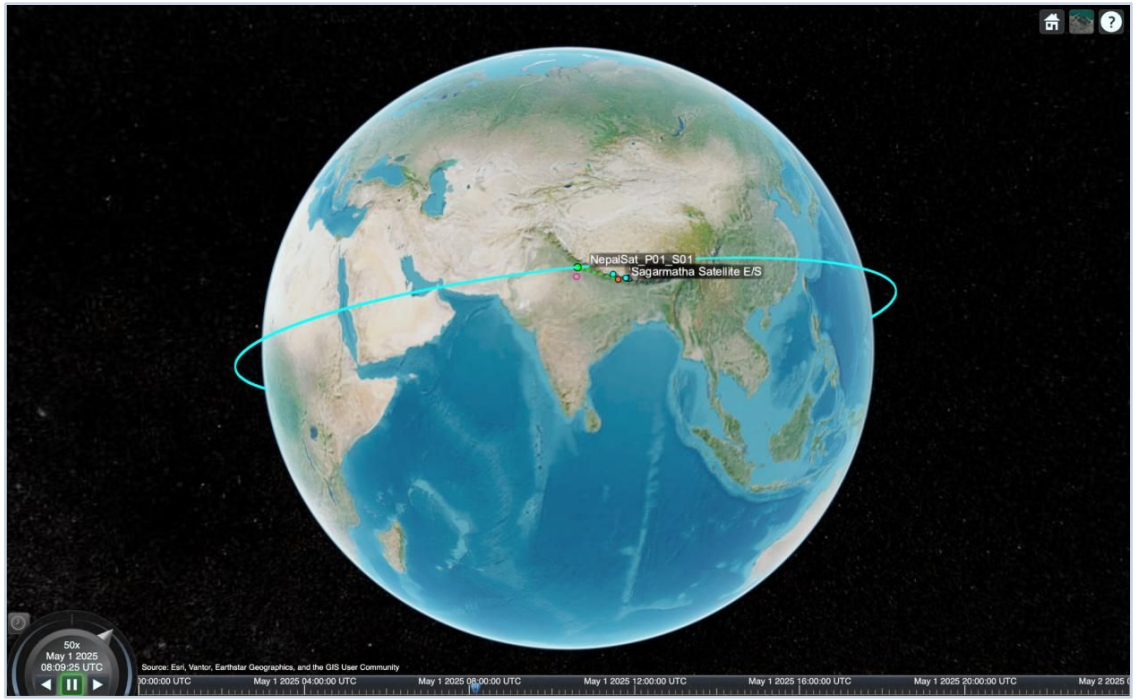


Figure 7: 3-D scenario viewer with sensor footprint and ground-station link

- A single satellite covers most of Nepal, but only in three daily passes.

Simulation and Result: Baseline Passes and Area Check

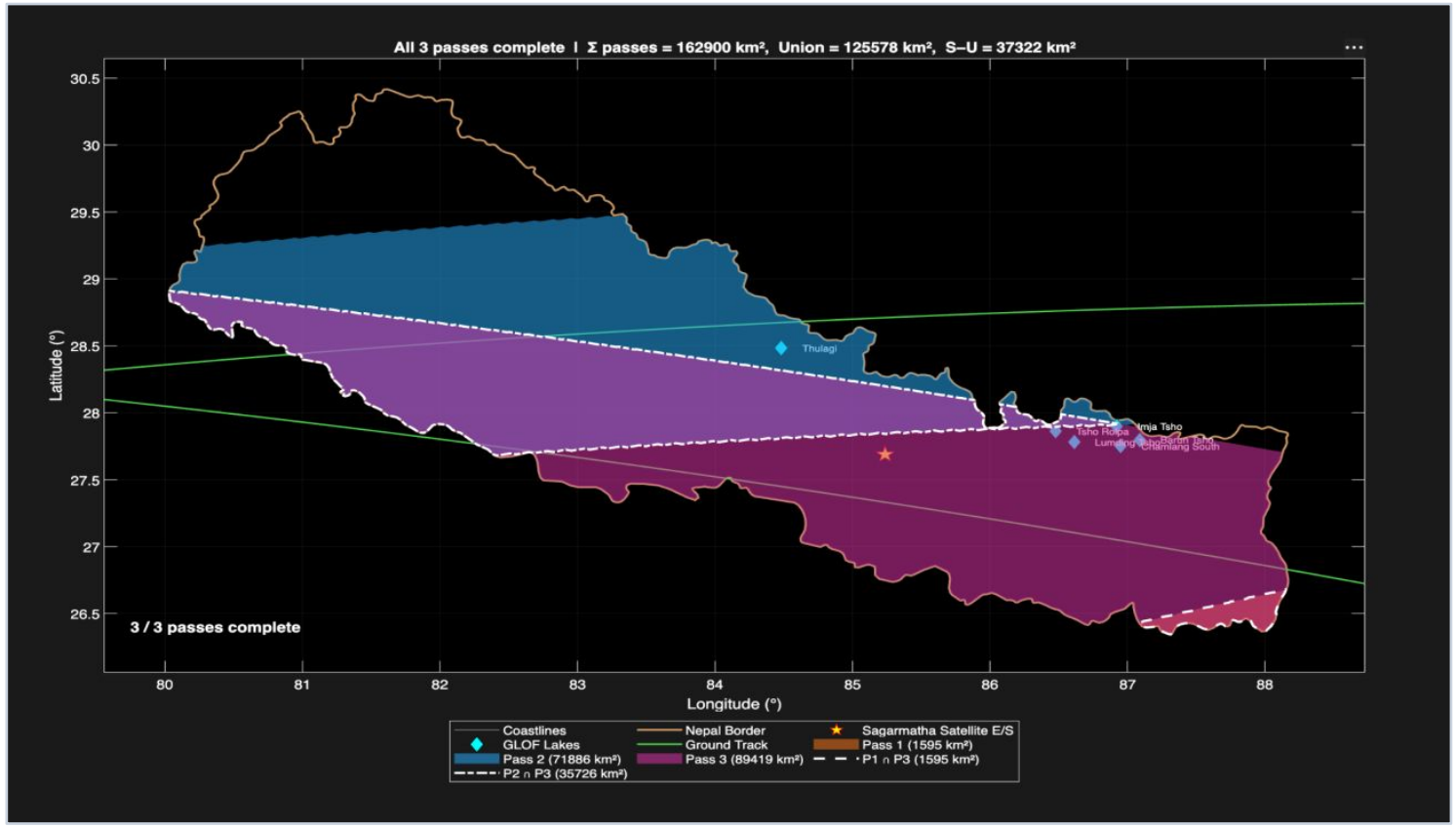


Figure 8: 2-D ground track over Nepal, three daily passes

1,595 km²

Pass 1

71,886 km²

Pass 2

89,419 km²

Pass 3

Union 125,578 km²

Simulation and Result: Per-Glacial-Lake Coverage

Lake	Passes	Dwell (min)	Covered
Imja Tsho	1	0.10	Yes
Tsho Rolpa	1	0.23	Yes
Thulagi	1	0.43	Yes
Chamlang South	1	0.27	Yes
Barun Tsho	1	0.20	Yes
Lumding Tsho	1	0.27	Yes

6 / 6

lakes covered by the baseline

6 / 6

lakes covered by the grown constellation

48 min

worst-case lake revisit

Simulation and Result: Step-Wise Constellation Growth

Step	Action	Incl. (°)	RAAN (°)	v (°)	Alt. (km)	Coverage	Revisit (min)	Wasserstein
1	Add satellite	28.662	345.45	215.12	-15.0	66.28%	49.50	0.070
2	Add satellite	28.662	345.45	332.35	+15.0	78.34%	30.71	0.044
3	**Add orbit**	**32.157**	**166.47**	**108.56**	**-35.1**	**85.91%**	**103.22**	**1.012**
4	Add satellite	32.157	166.47	28.84	+8.4	86.71%	80.42	0.765
5	Add satellite	32.157	166.47	312.69	-20.1	86.74%	65.07	0.607
6	**Add orbit**	**31.972**	**0.00**	**127.49**	**+34.6**	**92.56%**	**57.41**	**0.424**

+19.9% coverage

Step 1: also -51.0 min revisit, -51.5 min latency

+7.6% coverage

Step 3: first new plane, Wasserstein 1.01 (run high)

+5.8% coverage

Step 6: accepted despite a 7.7 min revisit regression

Simulation and Result: Final Grown Constellation

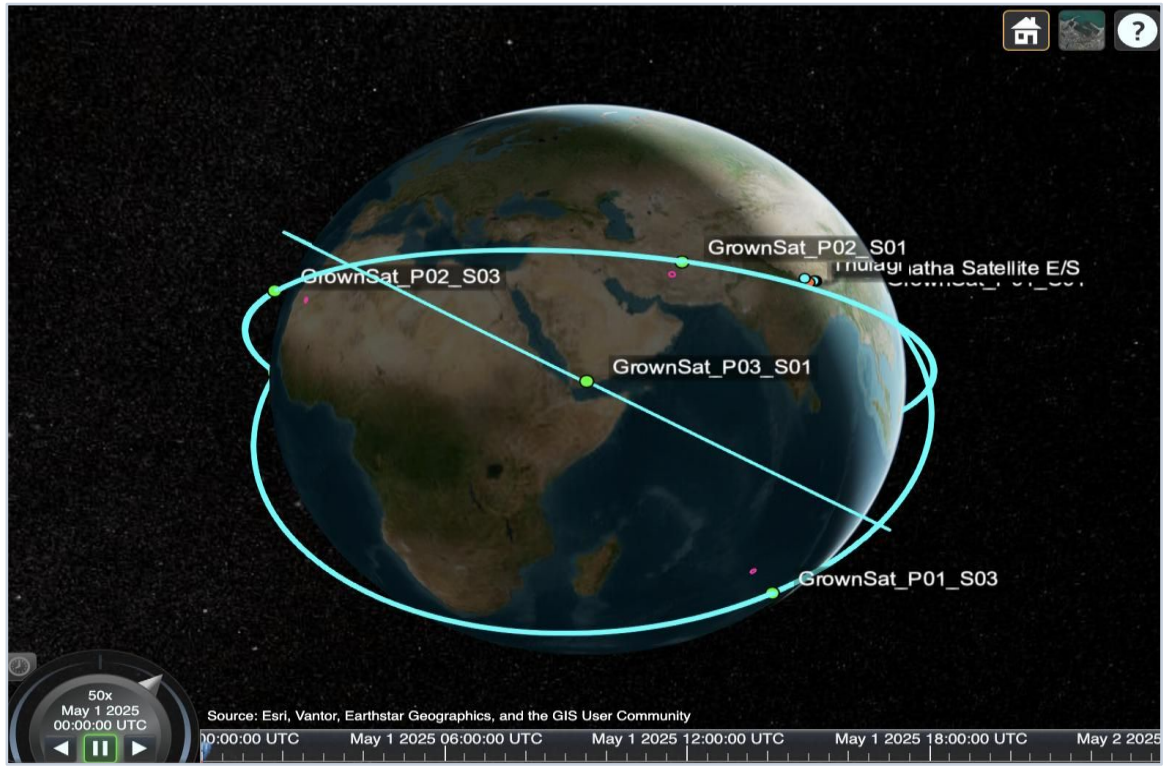


Figure 9: 3-D scenario viewer: 3 planes, 7 satellites

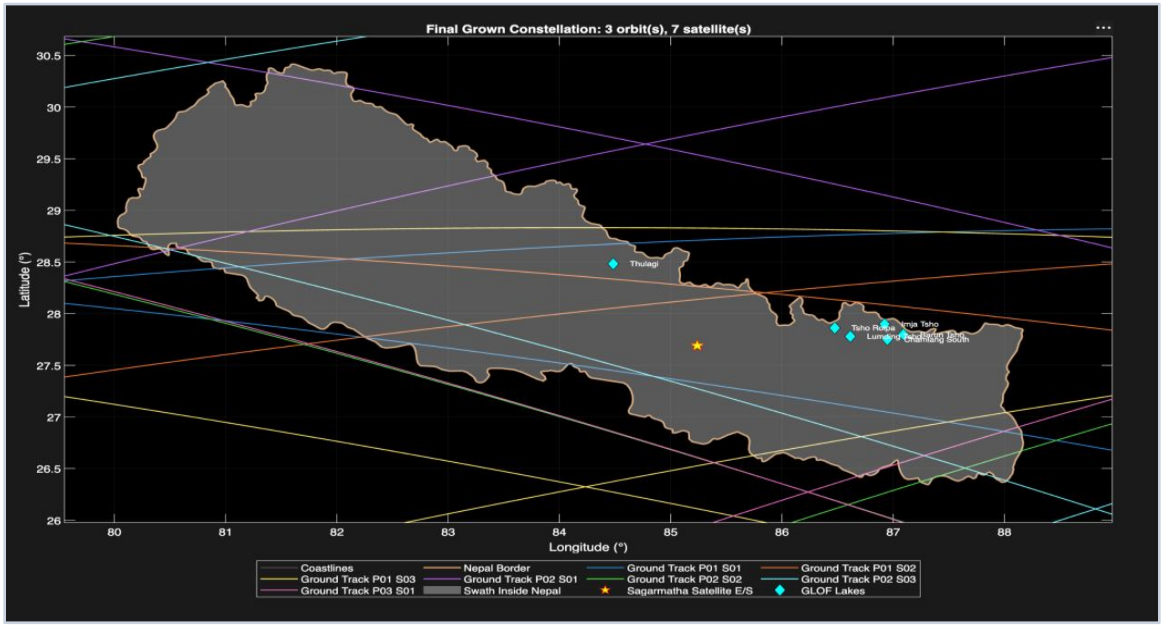


Figure 10: Static 2-D ground tracks over Nepal

7	3	18	92.56%
satellites	orbital planes	passes per day	Nepal coverage

Simulation and Result: Final Grown Constellation Metrics

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========= FINAL GROWN CONSTELLATION METRICS =========
Revisit time with swath:
  Average: 57.24 min
  Maximum: 215.00 min
Number of passes per day within swath: 18
SGS contact:
  Total contact time: 510.00 min
  Average per contact: 12.14 min
  Number of contacts: 42
Swath overlap time:
  Total: 10.00 min (41.67% of day)
Coverage fraction (accumulated area / Nepal area): 0.9625
Data latency (average time from observation to next SGS contact):
  Average: 35.56 min
Accumulated coverage over 24 h: 142022.39 km^2
Six lakes covered: 6/6; worst lake revisit: 31.00 min
Largest daily pass-time gap: 7.62 hr (baseline was 20.62 hr)
=====================================================
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Simulation and Result: Baseline vs Final Constellation

Metric	Baseline (1 sat)	Final (7 sats)
Average revisit (min)	**99.47**	**57.41**
Maximum revisit (min)	99.63	391.00
Passes per day	**3**	**18**
SGS contact (min)	77.33	523.00
Coverage of Nepal	85.11%	92.56%
Downlink latency (min)	**94.33**	**25.94**
Lakes covered	6 / 6	6 / 6

-42%

average revisit time

-72%

downlink latency

6×

passes per day

Simulation and Result: Temporal Gap Analysis

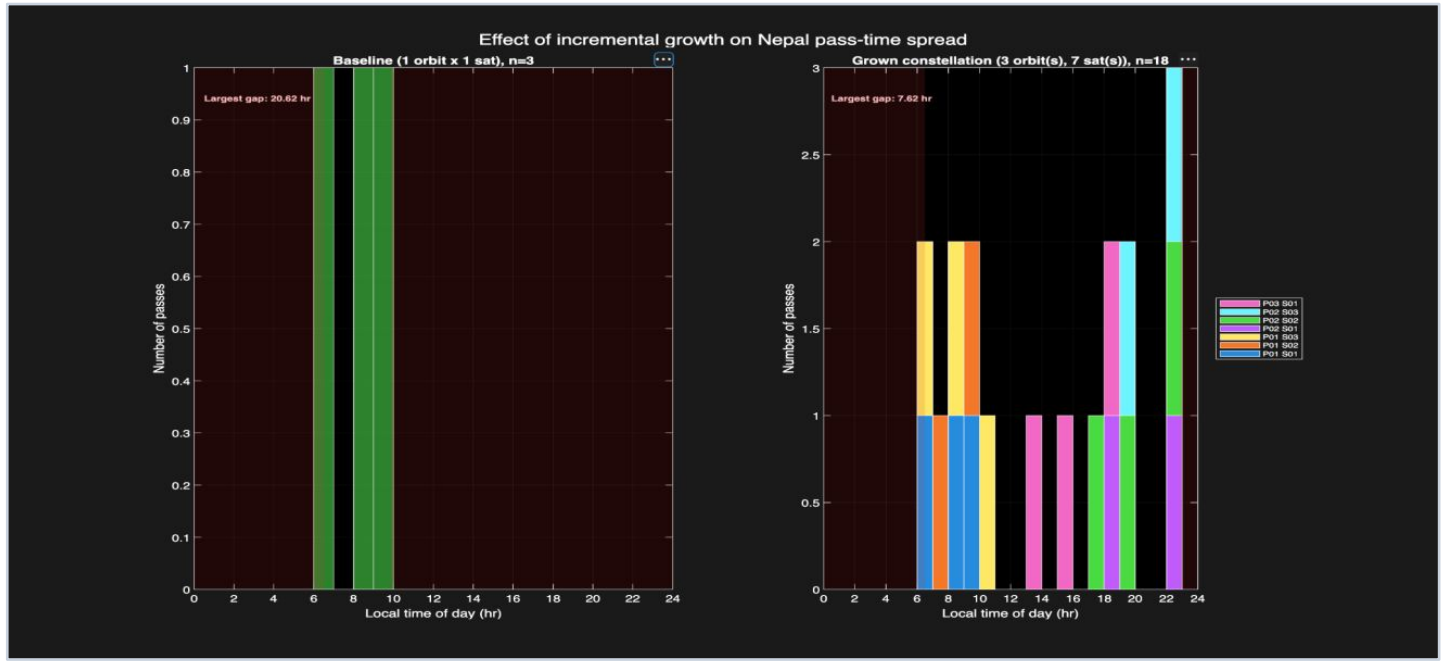


Figure 10: Pass-time-of-day histogram: baseline vs final

20.62 h

largest daily gap, baseline orbit

7.62 h

largest daily gap, final constellation

- Three pass windows: ~06-11, 13-16 and 17-23 h, one per plane; 00-05 h empty.
- Geometric cause, not a search failure: Inclination 28.7-32° is close to Nepal's 26-30°N band and same-plane satellites only shift passes inside their own window.
- Direct fix: More planes, with RAAN spaced around the day.

Gantt Chart

Phase	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Jan
Literature Review										
Requirement analysis and mission planning										
Orbital and coverage modelling										
Image extraction for Remote Monitoring										
Software pipeline integration										
Result verification and optimization										
Project Demonstration										

Work accomplished Work to be completed

Work Accomplishment

- Formulated single-satellite orbit selection as BO over inclination, RAAN and true anomaly using LHS seeding, GP surrogate and EI.
- Developed incremental constellation growth using same-plane and new-plane Bayesian Optimization with a hard worst-case lake-revisit constraint, growing the design to 7 satellites across 3 orbital planes.
- Extended the growth objective with Wasserstein temporal complementarity, largest-gap reduction, and gap-uniformity terms.

Work Accomplishment

- Implemented and validated coverage, revisit, ground-station, latency, and per-lake metrics, including independent swath-area verification.
- Produced 2-D ground-track maps, animated maps, a 3-D satellite viewer, and pass-time histograms with largest-gap annotation.
- Diagnosed the persistent daily coverage gap and used it to refine the growth objective.

Work Remaining

- Re-run the pipeline with the updated gap-uniformity objective and report full results, including per-lake performance.
- Reduce the residual coverage gap by testing more than 3 planes, higher-inclination planes, or explicit RAAN spacing.
- Extend analysis to multi-day and end-of-life conditions, including J2, drag, orbital decay, and station-keeping.
- Complete ground-station scheduling, downlink, and onboard sensing workflows for remote monitoring.

Work Remaining

- Integrate MATLAB orbital-design and Google Earth Engine remote-monitoring features.
- Validate results against representative TLE data.
- Provide detailed coverage, revisit time, and latency results for targeted GLOF sites.

Problem Encountered

- The 3-D viewer can time out during large scenario initialization.
- The standard revisit metric does not correctly account for observation gaps that cross midnight.
- Excessive largest-gap weighting initially reduced performance because same-plane searches could not shift coverage to a sufficiently different time of day.
- Full propagation and coverage simulation makes each Bayesian Optimization evaluation expensive.

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Thank You for Listening!